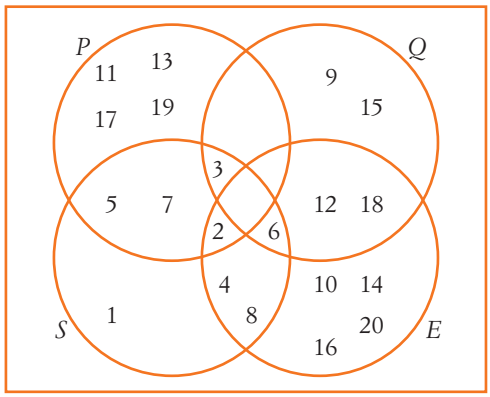
YEAR 11 MATHS METHODS INVESTIGATIVE TASK 2 SOLUTIONS

Probability

1 a



[5]

b i {2} **ii** {3, 6} **iii** {1} **iv** {3} **v** {1, 9, 15}

vi {1, 4, 8, 10, 14, 16, 20} **[6]**

c It must be composite, but not a multiple of 2 or 3 (and >8), so $5 \times 5 = 25$. **[1]**

d Example: $P \cap Q \cap \overline{S \cup E}$ or $P \cap Q \cap \overline{S}$. Yes, because there are no prime numbers which are multiples of 3 greater than 8. **[2]**

e Example: $P \cap Q$ or $P \cap E$. Yes, because 3 is the only one prime number which is a multiple of 3 (or the only even prime number). **[2]**

f i $\frac{4}{20} = \frac{1}{5}$ **ii** $\frac{9}{20}$ **iii** $\frac{10}{20} = \frac{1}{2}$ **[3]**

g i $\mathbf{P \cap S}$ **ii** $\mathbf{E \cap \overline{P}}$ **iii** $\mathbf{(E \cup S) \cap \overline{P}}$ **[3]**

2 a

	Boys		Girls		
	For	Against	For	Against	Totals
Year 10	6	111	10	123	250
Year 11	24	99	40	37	200
	30	210	50	160	450

[5]

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b i $\frac{240}{450} = \frac{8}{15} \approx 53\%$ **ii** $\frac{80}{450} = \frac{8}{45} \approx 18\%$

iii $\frac{16}{250} = \frac{8}{125} \approx 6\%$ **iv** $\frac{50}{450} = \frac{1}{9} \approx 11\%$

v $\frac{40}{450} = \frac{4}{45} \approx 9\%$ **vi** $\frac{40}{200} = \frac{1}{5} = 20\%$

vii $\frac{50}{80} = \frac{5}{8} \approx 63\%$ **viii** $\frac{99}{210} = \frac{33}{70} \approx 47\%$ **[8]**

c They are independent if:

$P(\text{Year 10 and for the proposal}) = P(\text{Year 10}) \times P(\text{For the proposal})$ **[1]**

$P(\text{Year 10 and for the proposal}) = \frac{16}{450} = \frac{8}{225}$ **[1]**

$P(\text{Year 10}) \times P(\text{For the proposal}) = \frac{250}{450} \times \frac{80}{450} = \frac{5}{9} \times \frac{8}{45} = \frac{8}{81} \neq \frac{8}{225}$ **[2]**

The events are not independent. **[1]**

3 a $2 \times 3 \times 2 = 12$ **[1]**

b Number of ways to choose main course and sweets $= 3 \times 2 = 6$ **[1]**

Probability $= \frac{1}{6}$. **[1]**

OR

Number of ways to choose a meal $= 2 \times 3 \times 2 = 12$

Number of these as described $= 2 \times 1 \times 1 = 2$ **[1]**

Probability $= \frac{2}{12} = \frac{1}{6}$. **[1]**

c Number of ways for two people $= 12 \times 12 = 144$ **[1]**

Choosing different entrees $= 2$ ways **[1]**

Choosing different mains $= 6$ ways (BS, BP, SB, SP, PB, PS) **[1]**

Choosing different desserts $= 2$ ways

Choosing differently $= 2 \times 6 \times 2 = 24$ **[1]**

Probability $= \frac{24}{144} = \frac{1}{6}$ **[1]**

d This excludes prawns and barramundi **[1]**

Number of meals $= 1 \times 2 \times 2 = 4$ **[1]**

e $\frac{1}{4}$ **[1]**

4 a There are 10 ways. Students should use some system, e.g. as below, '1 first, 2 first, 3 first, 4 first'.

114, 123, 132, 141, 213, 222, 231, 312, 321, 411 **[2]**

b $6 + 6 = 12$ **[1]**

c Let the radius of the small circle be r units.

Sample space area $= \pi(6r)^2 = 36\pi r^2$ **[2]**

Area of '6' circle $= \pi(r)^2 = \pi r^2 \text{—} P(6) = \dfrac{\pi r^2}{36\pi r^2} = \dfrac{1}{36}$ **[1]**

Area of '5' circle $= \pi(2r)^2 = 4\pi r^2 \text{—} P(5) = \dfrac{4\pi r^2 - \pi r^2}{36\pi r^2} = \dfrac{3\pi r^2}{36\pi r^2} = \dfrac{1}{12}$ **[1]**

Area of '4' circle $= \pi(3r)^2 = 9\pi r^2 \text{—} P(4) = \dfrac{9\pi r^2 - 4\pi r^2}{36\pi r^2} = \dfrac{5\pi r^2}{36\pi r^2} = \dfrac{5}{36}$

Area of '3' circle $= \pi(4r)^2 = 16\pi r^2 \text{—} P(3) = \dfrac{16\pi r^2 - 9\pi r^2}{36\pi r^2} = \dfrac{7\pi r^2}{36\pi r^2} = \dfrac{7}{36}$

Area of '2' circle $= \pi(5r)^2 = 25\pi r^2 \text{—} P(2) = \dfrac{25\pi r^2 - 16\pi r^2}{36\pi r^2} = \dfrac{9\pi r^2}{36\pi r^2} = \dfrac{1}{4}$

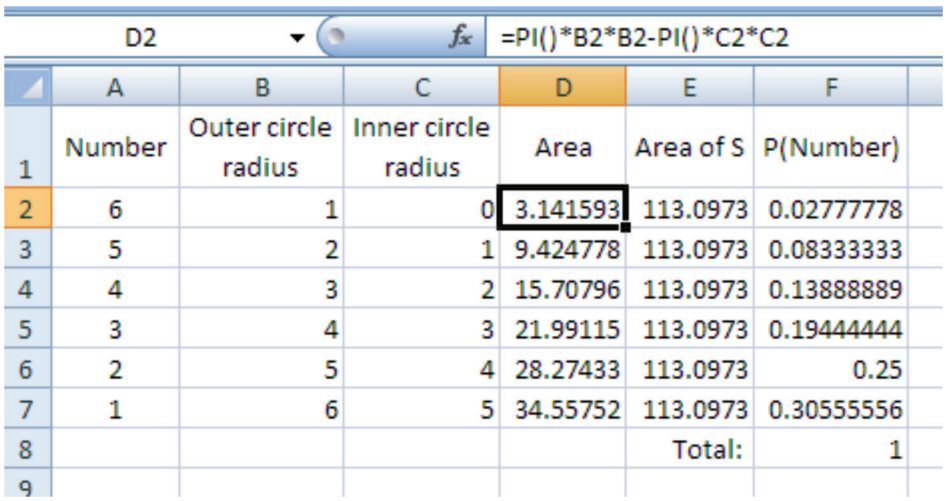
Area of '1' circle $= \pi(6r)^2 = 36\pi r^2 \text{—} P(1) = \dfrac{36\pi r^2 - 25\pi r^2}{36\pi r^2} = \dfrac{11\pi r^2}{36\pi r^2} = \dfrac{11}{36}$ **[2]**

Table formed by converting the fractions to decimals: **[2]**

Score	P(score)
6	0.03
5	0.08
4	0.14
3	0.19
2	0.25
1	0.31

d Sample spreadsheet with the formula for D2. Drag down where appropriate.

[5]



	A	B	C	D	E	F
1	Number	Outer circle radius	Inner circle radius	Area	Area of S	P(Number)
2	6	1	0	3.141593	113.0973	0.02777778
3	5	2	1	9.424778	113.0973	0.08333333
4	4	3	2	15.70796	113.0973	0.13888889
5	3	4	3	21.99115	113.0973	0.19444444
6	2	5	4	28.27433	113.0973	0.25
7	1	6	5	34.55752	113.0973	0.30555556
8					Total:	1
9						

E2: = PI()*36 then drag down.

F2: = D2/E2 then drag down.

There are other ways of doing the spreadsheet.

e It is strongly recommended students use calculators which are able to manipulate fractions for this part.

i $\frac{1}{4} \times \frac{1}{12} \times \frac{1}{36} = \frac{1}{1728}$ **[2]**

ii $\frac{1}{36} \times \frac{1}{36} \times \frac{1}{36} = \frac{1}{46\,656}$ **[1]**

iii $1 - \frac{1}{46\,656} = \frac{46\,655}{46\,656}$ **[1]**

iv Students should use the results from part **a**. There are three ways with two 1s and a 4, all with the same probability. Likewise six ways with a 1, a 2 and a 3, and one way of three 2's (222). **[2]**

Two 1s and a 4: $3 \times \frac{11}{36} \times \frac{5}{36} \times \frac{5}{36} = \frac{275}{15\,552}$ **[2]**

A 1, a 2 and a 3: $6 \times \frac{11}{36} \times \frac{1}{4} \times \frac{7}{36} = \frac{77}{864}$ **[1]**

222: $\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} = \frac{1}{64}$ **[1]**

$P(\text{Total is 6}) = \frac{275}{15\,552} + \frac{77}{864} + \frac{1}{64} = \frac{119}{972}$ **[1]**

f i No as proportional to r^2 so the pattern would be the same. **[1]**

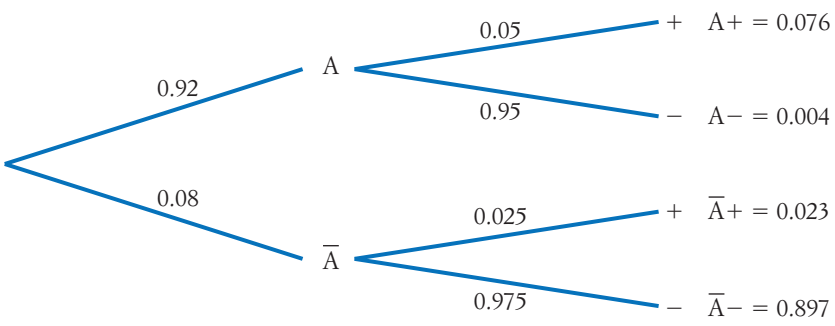
ii $P(10) = \frac{1}{100}$, $P(9) = \frac{3}{100}$, $P(8) = \frac{5}{100} = \frac{1}{20}$, $P(7) = \frac{7}{100}$,
$P(6) = \frac{9}{100}$, $P(5) = \frac{11}{100}$, $P(4) = \frac{13}{100}$, $P(3) = \frac{15}{100} = \frac{3}{20}$,
$P(2) = \frac{17}{100}$, $P(1) = \frac{19}{100}$ **[2]**

iii $P(n) = \frac{1}{n^2}$, $P(n-1) = \frac{3}{n^2}$, $P(n-2) = \frac{5}{n^2}$, … $P(1) = \frac{2n-1}{n^2}$ **[2]**

iv The sum of the numerators will be the sum of the first n odd numbers and must equal n^2 since the sum of the probabilities must be 1. **[2]**

The sum of the first n odd numbers must be n^2. **[1]**

5 a



[5]

b A person will test negative and not have the disease. **[1]**

c i $0.076\ (\mathbf{A}+)$ **[1]**

ii $0.076 + 0.023 = 0.099$ **[1]**

iii $0.004 + 0.023 = 0.027$ **[1]**

iv $\dfrac{P(\bar{\mathbf{A}} \cap +)}{P(+)} = \dfrac{0.023}{0.023 + 0.076} = \dfrac{23}{99} \approx 0.23$ **[2]**

d $\dfrac{23}{99} \times 400 \approx 93$ **[1]**

This is a large proportion, due to the fact that most people don't have the disease. It could be a problem if treatment of the disease was hazardous to healthy people.

Other comments are possible. **[2]**

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